

ZACHARY KING

zacharygking@icloud.com • [linkedin.com/in/zacharygking](https://www.linkedin.com/in/zacharygking) • github.com/zacharygking

Applied ML at Apple, 8 years. Authored a team's multi-label evaluation-metric standard, earned named project responsibility, and then owned multilingual LLM features that shipped. Took over the Professional tone's internationalization and adapter training across 20+ locales in Writing Tools. Then end-to-end owner of another text feature.

EXPERIENCE

Apple Inc.

2018 – Present

Applied ML Engineer

2019 – Present

Input Experience – NLP • Focus: *LLM adapters, evaluation and prompt optimization*

2024 – Present

- **Multilingual tone adapters:** Owned the internationalization of the Professional tone. Built the data-synthesis pipelines, fine-tuned the LoRA adapters on locale data produced with localization partners, and took them through sign-off to release. Shipped in 20+ locales over two waves. Separately moved the Friendly tone onto new base models.
- **End-to-end feature ownership:** Owned a Writing Tools feature through prototyping, data synthesis and evaluation. Built its per-locale prompt-optimization loop on the team's platform (DSPy/GEPA), and gave each quality judge its own reward so the optimizer could tell a near miss from an outright failure. The feature reached its shipping evaluation targets.
- **Ground-truth benchmarking and adjudication:** Built the evaluation datasets, benchmarked the automated grader against human grades, and adjudicated every disagreement. The grader proved more defensible than the human baseline, and most of the rest traced to unreliable human labels.

Input Experience – NLP • Focus: *Multilingual keyboard language models*

2023 – 2024

- **Handwriting language models:** Extended the character n-gram pipeline to 13 locales, and shipped its models behind Scribble, Live Text, Smart Script and Visual Intelligence. The n-grams compressed about 290x on device.
- **Czech and Italian keyboard models:** Shipped both, integrating the per-locale models plus their lexicon and unigram pipelines into autocorrect, predictive typing and QuickPath, and cleared release-blocking swipe defects.
- **Privacy-preserving vocabulary learning:** Used differentially private on-device statistics to pull out-of-lexicon terms and typed names into the keyboard lexicon. Won privacy and legal approval at a leadership review and shipped on 5 platforms.
- **Multilingual shadow evaluation:** Set the shadow-evaluation methodology for a new on-device model across nine languages, bringing six independent per-language owners onto one protocol. Documented the platform gap that left two languages unevaluable rather than reporting seven as evaluated.

Proactive Intelligence • Focus: *On-device intelligence platform*

2019 – 2023

- **Multilingual index:** Delivered the full Spanish index, then wrote the multilingual build, with parallel per-language builds across 6 languages, selectable combinations and automated asset upload. Later decoupled its data from language-specific builds, cutting about 85% of the asset weight from OS images.
- **Neural topic classifier:** Authored the dataset spec, the architecture proposal and the team's multi-label evaluation-metric standard, then was named the engineering owner about ten months later to replace the shipped Lucene/Java topic classifier with a neural one, sized to the Neural Engine's on-device memory budget.
- **Safari Tab Group naming:** Mapped a web-directory taxonomy onto knowledge-graph entities so per-page topics rolled up into higher-level topics for grouping. Wrote the naming UI and the topic lookup, and shipped suggested Tab Group names on iOS and macOS.

Software Engineer, Siri (rotation program, six three-month projects)

2018 – 2019

- **Siri Knowledge Graph** (two rotations): Automated knowledge-graph gap detection routing candidate facts to crowd verification, and reworked clustering and indexing so annotators could clean multilingual training sets.
- **Proactive Intelligence:** Stood up a search-quality harness across 4 retrieval backends behind one rating UI.
- **Siri Domains:** Shipped NBA Draft answers to iOS, macOS, watchOS and tvOS.
- **Core ML and Siri Visual Intelligence:** Prototyped a Core ML profiler in Xcode exposing per-layer latency and compute-unit placement, and extended an on-device text classifier to product categories for visual search.

SKILLS

Machine learning & evaluation: LLM-as-judge evaluation, benchmark and rubric design, human evaluation protocols, prompt optimization (DSPy/GEPA), LoRA/PEFT fine-tuning, NLP, on-device model compression

Engineering: Python, Java, Swift, Objective-C, C, Go; PyTorch, Core ML, pandas, Lucene; distributed GPU training, CI/CD

PATENTS *Systems and Methods for Proactively Identifying and Providing an Internet Link on an Electronic Device.* US Patent 11,620,033 B2, co-inventor, Apple Inc., 2023.

EDUCATION

B.S. Computer Science, University of California, Los Angeles, 2018

Software Engineering Intern: **Google** (YouTube abuse engineering, 2017), **Box** (2016), **IDMLoco** (2015)